Problem Set 5

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: CO A.3–A.5; MML 5.1–5.2. Problems marked **[code]** should be submitted as a program in a language of your choice (Julia or Python recommended). The cited exercise numbers are from MML (Deisenroth et al., 2021).

1. MML, Chapter 5 exercises: any three on partial derivatives, gradients, and the Hessian (e.g. Exercises 5.1–5.4).
2. For the quadratic form $f(x) = x^\top A x$ with A symmetric, derive the gradient ∇f and the Hessian $\nabla^2 f$ by hand, and explain how the definiteness of A (via its eigenvalues) determines whether f has a minimum, a maximum, or a saddle at the origin.
3. **[code]** For $f(x, y) = x^2 + 2xy + 3y^2$, compute the gradient and Hessian by hand, then verify both numerically by finite differences at the point $(1, 1)$. Plot the contours of f.
4. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).